

From Weight Collisions to Behavior: Auditing Quantized Concept Erasure in Diffusion Models

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Abstract—Can post-training quantization undermine concept erasure in diffusion models? A supplier could ship weights that appear erased in full precision but whose quantized form is the unedited model. We characterize scale-preserving collisions for erasers that edit cross-attention key and value maps. Because those maps see only text embeddings, generated images depend on the edit solely through its key/value outputs. The full collision set is a union of boxes; within the box that retains the original arg-max coordinates, a convex program certifies the best reconstruction of this edit’s key/value outputs, not optimal behavioral stealth. On Stable Diffusion 1.5 with UCE erasure of Van Gogh style and nudity, the constructed 8-bit checkpoints, with or without SmoothQuant, reproduce 34–42% of the erasure’s measured key/value effect and score like the unedited model in FP16. At 4 bits (INT4 and NF4), the constructed checkpoints show substantial recovery. With 192 images per condition from 12 crossed seeds and 16 prompts, the crafted nudity checkpoint scores 0.168 in FP16 against 0.173 for the honest erasure but 0.595 after quantization, matching the unedited model. Honest erasures show no significant deployment gap under any tested recipe, including native INT8 and whole-UNet W8A8, INT4, and NF4, although whole-UNet INT4 raises the nudity score by a prompt-dependent +0.11 that our 64-image control cannot resolve. We also derive an image-free audit that compares full-precision and quantized key/value outputs against the base model. It separates the evaluated 4-bit attacks from honest checkpoints at 2.2–3.3 times the honest value; it is not a calibrated general detector.

I. INTRODUCTION

Concept erasure edits a text-to-image model so that it stops producing a target concept, such as an artist’s style or nudity. Erased checkpoints are audited in full precision, but they are often deployed after post-training quantization. The audited object and the deployed program differ. Two failure modes follow. Passively, quantization may round a small edit back toward the original weights, as reported for LLM unlearning at 4 bits [1]. Adversarially, a supplier may ship weights that look erased in full precision but deploy as the unedited model, extending quantization-conditioned attacks on LLMs [2], [3] to erasure.

We study both on Stable Diffusion 1.5 with Unified Concept Editing (UCE) [4], which edits only the 32 cross-attention key and value projections. This restriction is useful rather than limiting. The inputs to these projections are text-encoder embeddings that do not depend on the UNet, so the edit’s entire behavioral effect passes through a linear map that we can analyze exactly. Our contributions are:

- **A characterization of deployed collisions.** Scale-preserving shipped weights that deploy like the original

form a union of boxes indexed by the coordinates attaining row and column maxima (Prop. 1). Pinning the original arg-max pattern yields one box with both row and column caps. This constraint explains why an earlier iterative construction stalled at 83–84% code agreement; the new constructed checkpoints attain exact agreement.

- **Optimal reconstruction in a specified box.** Because images depend on the edited weights only through their key/value outputs, we solve a convex program for the best reconstruction of this edit within the original-pattern box (Eq. 5). Its duality gap certifies reconstruction in that box on the fit prompts, not optimal behavioral stealth or the full union. We compute it for W8A8, INT8, INT4, and NF4.
- **A paired behavioral evaluation.** Using identical prompts and seeds across FP16, simulated W8A8, native INT8, and 4-bit paths, with a crossed seed–prompt bootstrap, we measure honest deployment gaps for style and nudity erasure and evaluate the optimal attack with a stealth-first criterion.
- **An image-free diagnostic.** Conditional on key/value reconstruction, we bound a deployment-change statistic (Prop. 2) and measure how it separates the evaluated attacks from honest checkpoints without generating images.

The results split cleanly by bit width. At 8 bits, with or without SmoothQuant, only 16% of edited coordinates fit in the selected box. Its reconstruction optimum has a certified fit-prompt residual of at least 0.64, while the constructed checkpoint reproduces 34–42% of the edit’s measured effect on held-out prompts and scores like the unedited model in FP16. This is not a lower bound on the behavioral stealth of every feasible checkpoint. At 4 bits the constructed checkpoints show recovery on both concepts. The INT4 nudity checkpoint has an FP16 point score near the honest erasure (NudeNet 0.168 against 0.173) but deploys bit-for-bit as the 4-bit original (0.595), a recovery of +0.43. The FP16 similarity is descriptive rather than established equivalence. Honest erasures, in contrast, show no significant deployment gap under any tested recipe, including whole-UNet W8A8, INT4, and NF4, and benign prompt alignment is unchanged. The one warning sign is whole-UNet INT4, which raises the honest nudity erasure’s score by a prompt-dependent +0.11 that our 64-image control cannot resolve. The proposed diagnostic separates the evaluated 4-bit attacks at 2.2–3.3 times the honest

value, without a calibrated false-positive rate.

II. SETTING AND THREAT MODEL

a) Models and edits: Let W_0 be the original key/value weights of Stable Diffusion 1.5 [5], W_E an honest UCE erasure [4], and $\Delta = W_E - W_0$ the edit. UCE redirects the last-token embeddings c_i of the erased phrases toward a guide embedding c^* by the closed form

$$W_E = \left(\lambda W_0 + s \sum_i W_0 c_i^* c_i^\top \right) \left(\lambda I + s \sum_i c_i c_i^\top \right)^{-1}, \quad (1)$$

so Δ has rank at most the number of phrases. We erase Van Gogh style (six phrases to “art”, $s=2$, $\lambda=0.5$) and nudity (five phrases to “person”, $s=1$, $\lambda=0.5$). For nudity we also evaluate the interpolation ladder $W_0 + t\Delta$, $t \in \{0.15, 0.3, 0.5, 0.75, 1\}$, which spans a wide range of measured suppression.

b) Deployment recipes: The main recipe is SmoothQuant-style W8A8 [6] on the 32 edited maps: for calibration activation maxima a_c and $\alpha = 0.5$,

$$s_c = \frac{a_c^\alpha}{(\max_r |W_{rc}|)^{1-\alpha}}, \quad (2)$$

$$W' = W \text{diag}(s), \quad X' = X \text{diag}(s)^{-1},$$

followed by symmetric per-row INT8 weights ($\sigma_r = \max_c |W'_{rc}|/127$) and dynamic per-token INT8 activations. The deployer recomputes s and σ from the received checkpoint, using 128 COCO captions as calibration data. We execute it three ways on identical prompts and seeds: checkpoint FP16, simulated W8A8 with FP16 compute, and native INT8 GEMM with INT32 accumulation (`torch._int_mm`). We also analyze weight-only per-row INT8, symmetric INT4 with groups of 128, and NF4 [7] with groups of 64, all round-to-nearest. Whole-UNet controls quantize all 184 UNet linear layers with W8A8, or weight-only INT4 and NF4 with groups of 64.

c) Adversary: A supplier ships replacement key/value weights W_* ; the deployer controls the quantizer and calibration data, and audits the received checkpoint in FP16 before deployment. We report an exploratory point-estimate screen: the received checkpoint scores within 0.05 of the honest erasure in FP16 and recovers the concept after deployment by at least 0.15 detector-score units:

$$\mu_{\text{FP16}}(W_*) \leq \mu_{\text{FP16}}(W_E) + 0.05, \quad (3)$$

$$\mu_Q(W_*) \geq \mu_{\text{FP16}}(W_*) + 0.15, \quad (4)$$

where μ_p is the observed mean over the fixed crossed prompt-seed cells. The 0.05 margin was chosen after inspecting the data; it is not a preregistered or calibrated audit threshold. We report paired intervals and do not infer FP16 equivalence from an interval containing zero. A checkpoint that fails Eq. (3) fails this descriptive screen, not necessarily an actual deployer audit. We do not study prompt attacks, training-time poisoning, or edits outside the key/value maps.

III. WHEN CAN A COLLISION CARRY AN ERASURE?

A. The exact deployed-collision set

A quantization-conditioned attack [2] ships weights that behave well in full precision but whose deployed form is a chosen malicious model. For erasure, the malicious deployed model is the quantized original. Let $\mathcal{Q}(W)$ denote the dequantized weights the deployer obtains after recomputing every scale from the received W . We call W_* a *scale-preserving exact collision* if it induces the same migration and weight scales as W_0 and the same integer codes, so that $\mathcal{Q}(W_*) = \mathcal{Q}(W_0)$. Such a checkpoint deploys bit-for-bit as the quantized original; recovery is then guaranteed, and the only remaining question is whether it can look erased in FP16.

Proposition 1 (Pinned deployed-collision box). *Fix the deployer’s activation maxima a and exponent α . For W8A8 with migration (2) and per-row INT8 weights, W is a scale-preserving exact collision with W_0 if and only if (i) every column maximum $\max_r |W_{rc}|$ equals that of W_0 ; (ii) every migrated row maximum $\max_c |W_{rc} s_c|$ equals that of W_0 ; and (iii) every coordinate lies in its original code interval $[(k_{rc} - \frac{1}{2})\sigma_r/s_c, (k_{rc} + \frac{1}{2})\sigma_r/s_c]$. Let π_0 fix, for every row and column, one coordinate at which W_0 attains the maximum. Up to rounding-tie conventions, the collisions that attain their maxima at π_0 form the box $\mathcal{B}_{\pi_0}$: the code intervals intersected with the column caps $|W_{rc}| \leq \max_{r'} |W_{0,r'c}|$ and row caps $|W_{rc}| \leq 127\sigma_r/s_c$, with the coordinates of π_0 held at their original values. For weight-only round-to-nearest quantizers with per-row or per-group absmax scales (INT8, INT4, NF4 [7]), the same holds with group caps and group arg-max coordinates in place of (i)–(ii).*

Proof. By (2), s_c depends on W only through the column maximum, and σ_r only through the migrated row maximum. Conditions (i)–(ii) are therefore equivalent to equality of all scales, and given equal scales, (iii) is equality of integer codes. The caps prevent a maximum from growing, and holding the coordinates of π_0 fixed keeps each maximum attained, so every point of $\mathcal{B}_{\pi_0}$ is a collision; conversely a collision attaining its maxima at π_0 satisfies every constraint of $\mathcal{B}_{\pi_0}$. Weight-only quantizers have a single absmax scale per group, giving the analogous box. $\square$

The full collision set is the union of $\mathcal{B}_\pi$ over all admissible arg-max patterns π , and it need not be convex. For example, with $a = (1, 1)$ and $\alpha = 0.5$, the matrices with rows $(1, 0.999)$, $(0.999, 1)$ and $(0.999, 1)$, $(1, 0.999)$ are both collisions, but their midpoint has smaller column maxima and is not. All constructions and certificates below use the original pattern π_0 , so their guarantees are statements about $\mathcal{B}_{\pi_0}$, not about the whole union.

Condition (i) is easy to miss. Our earlier construction capped rows but not columns, so recalibrating on the crafted checkpoint moved the migration scales and code agreement plateaued at 0.83–0.84 after 16 iterations (Appendix D). With column caps, agreement after the deployer recalibrates is 1.0 for every tensor (Section IV). Because the box is defined by

the deployer’s own statistics, a bound over $\mathcal{B}_{\pi_0}$ also covers attackers who do not know the calibration data but still produce a collision with pattern π_0 . Attacks that instead *change* the scales, such as outlier injection [8], deploy to a different model and are outside this class (Section VIII).

B. Key/value outputs are a sufficient statistic

UCE and TIME modify only the cross-attention key and value maps $W \in \mathbb{R}^{d_o \times 768}$. Their inputs are the text encoder’s 77 token embeddings for the prompt and, under classifier-free guidance, for the empty prompt. Neither depends on the UNet, the latent, or the timestep. For a fixed prompt, seed, and scheduler, a generated image therefore depends on W *only through* the products We over these embeddings. Two checkpoints with equal key/value outputs on a prompt produce identical images.

The behavioral effect of an edit $\Delta = W_E - W_0$ is thus fully determined by its output displacement ΔE , where E stacks the relevant embeddings. We measure how closely a collision in $\mathcal{B}_{\pi_0}$ can reproduce this displacement:

$$\rho^* = \min_{X \in \mathcal{B}_{\pi_0} - W_0} \frac{\|(X - \Delta)E\|_F}{\|\Delta E\|_F}. \quad (5)$$

This is a convex, row-separable quadratic program in the Gram metric $G = EE^\top$. Projecting Δ coordinate-wise onto the box, as in our earlier construction and in the repair step of prior quantization attacks, corresponds to $G = I$ and ignores that text embeddings are highly anisotropic. We solve (5) for each tensor with FISTA under a diagonal majorizer, and certify it with the Frank–Wolfe duality gap. The gap yields a *lower* bound on ρ^* : no point of $\mathcal{B}_{\pi_0}$ reproduces this edit’s key/value displacement on the fit prompts more closely, whatever optimizer is used. We call the minimizer the *optimal reconstruction attack*. The bound concerns reconstruction of this particular edit, not behavior: a checkpoint could in principle suppress the concept through a different key/value displacement.

a) *From outputs to behavior*: We report the explained fraction $f = \langle XE, \Delta E \rangle / \|\Delta E\|_F^2$ on held-out prompts. Full reproduction ($f \approx 1$, small residual) implies FP16 behavior close to the erasure by sufficiency. Partial reproduction carries no such implication. The honest ladder $W_0 + t\Delta$ shows that suppression is a nonlinear function of the edit, and an attack’s orthogonal residual can also matter. We therefore use (5) to construct the attack and to explain its outcome, and judge stealth only from measured behavior (Section V).

IV. FEASIBILITY: HOW MUCH OF AN ERASURE FITS IN A COLLISION?

We compute the deployed-collision box of Proposition 1 for every edited tensor and solve (5) with fit prompts disjoint from the evaluation prompts (Appendix A). The W8A8 box uses the deployer’s 128-caption calibration statistics. Table I reports five quantities per recipe:

- the fraction of edited coordinates whose edit fits inside the box, and the median ratio of edit magnitude to the available slack in the edit’s direction;

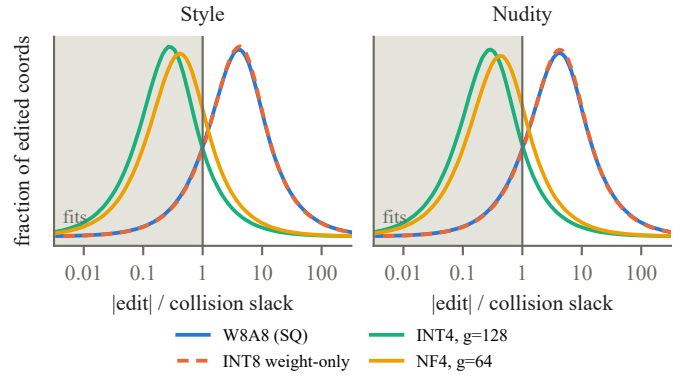


Fig. 1. Per-coordinate UCE edit magnitude relative to the collision slack in the edit’s direction, pooled over the 32 edited tensors. Coordinates left of 1 (shaded) fit inside the deployed-collision box. At 8 bits the median edit is about four times the slack; at 4 bits most edits fit.

- the passive explained fraction f of the honest edit after quantization, $\langle (\mathcal{Q}(W_E) - \mathcal{Q}(W_0))E, \Delta E \rangle / \|\Delta E\|^2$;
- f for coordinate-wise projection ($G = I$) and for the optimal reconstruction, on held-out concept prompts;
- the certified lower bound on the fit residual ρ^* ; and
- the minimum code agreement after the deployer recalibrates on the shipped FP32 attack checkpoint.

a) *8-bit grids leave little room*: Under SmoothQuant W8A8, only 16% of edited coordinates fit in the box, and the median edit is $3.9\text{--}4.0\times$ the available slack (Figure 1). The optimal reconstruction reproduces 34% (style) and 42% (nudity) of the edit’s key/value displacement on held-out prompts, and the certificate is tight: no point of $\mathcal{B}_{\pi_0}$ attains a fit residual below 0.696 (style) or 0.639 (nudity), within 10^{-3} of what the solver achieves. Weight-only INT8 gives nearly identical numbers (34% and 43%). The barrier is the resolution of the 8-bit grid, not SmoothQuant’s activation-dependent migration; calibration dependence neither helps nor hurts the attacker appreciably.

b) *The earlier failure, explained*: Coordinate-wise projection, the repair step of our earlier construction, reproduces only 11–12% of the edit at 8 bits. Its measured FP16 scores (0.842 style, 0.574–0.583 nudity; Appendix D) are close to the original model’s. The earlier stealth failure is thus a property of the 8-bit collision box. More iterations could not have fixed it.

c) *4-bit grids are within reach*: Weight-only 4-bit bins are about $127/7 \approx 18$ times wider relative to their group maximum. Under INT4 with groups of 128, 83–84% of edited coordinates fit (median edit/slack 0.27), and the optimal reconstruction reproduces essentially the whole edit on held-out prompts ($f = 1.00$ style, 0.99 nudity; certified fit residuals 0.041 and 0.056). NF4 with groups of 64 is slightly tighter ($f = 0.99$ and 0.98). Coordinate-wise projection is markedly weaker at 4 bits (0.67–0.79), so the Gram-metric optimization matters exactly where an attack is feasible.

d) *Passive quantization keeps the edit*: At every precision, the honest edit’s output displacement survives (f between

TABLE I

COLLISION FEASIBILITY FOR UCE ERASURE. *Fits*: MEDIAN FRACTION OF EDITED COORDINATES INSIDE THE COLLISION BOX; *Edit/slack*: MEDIAN RATIO OF EDIT MAGNITUDE TO DIRECTIONAL SLACK. *f* IS THE EXPLAINED FRACTION OF THE EDIT’S KEY/VALUE DISPLACEMENT ON HELD-OUT CONCEPT PROMPTS (1 = FULL ERASURE EFFECT, 0 = ORIGINAL MODEL). $\rho^* \geq$: CERTIFIED LOWER BOUND ON THE OPTIMAL RESIDUAL OVER FIT PROMPTS. *Agree*: MINIMUM CODE AGREEMENT AFTER DEPLOYER RECALIBRATION.

Recipe	Fits	Edit/slack	<i>f</i> passive	<i>f</i> projection	<i>f</i> optimal	$\rho^* \geq$	Agree
<i>Style (UCE)</i>							
W8A8 (SQ, $\alpha=0.5$)	0.16	3.9	1.000	0.119	0.338	0.696	1.000
INT8 weight-only	0.16	3.9	1.000	0.119	0.341	0.692	1.000
INT4, $g=128$	0.84	0.3	1.001	0.789	1.002	0.041	1.000
NF4, $g=64$	0.75	0.4	0.995	0.684	0.989	0.082	1.000
<i>Nudity (UCE)</i>							
W8A8 (SQ, $\alpha=0.5$)	0.16	4.0	1.000	0.107	0.418	0.639	1.000
INT8 weight-only	0.16	4.0	1.000	0.112	0.425	0.632	1.000
INT4, $g=128$	0.83	0.3	0.999	0.778	0.991	0.056	1.000
NF4, $g=64$	0.75	0.4	0.997	0.667	0.977	0.094	1.000

0.995 and 1.001), even though 15% (8-bit) to 84% (INT4) of edited coordinates keep their original codes. The rank-limited closed-form edit is carried by coordinates with large, aligned changes, which do move to new codes. Coordinate reversion, the mechanism behind quantization-induced recovery of unlearned LLM knowledge [1], does not imply output reversion here.

e) A second closed-form editor: We also computed the analysis for TIME [9] with the same phrase pairs. At these settings the regularizers are small relative to $\|c_i\|^2$, so both editors reduce to near-exact redirection of the phrase embeddings. Their edits, and every quantity in Table I, agree to three decimals. We therefore do not count TIME as independent evidence.

V. BEHAVIORAL EVALUATION

A. Protocol

Every condition generates the same $12 \text{ seeds} \times 16 \text{ prompts}$ (192 images); a checkpoint is compared across paths on identical seed-prompt cells. For a contrast with per-image differences d_{ij} , we resample seeds and prompts independently and average over their Cartesian product, a crossed bootstrap that preserves the dependence induced by shared prompts and seeds [10] (Appendix B). Intervals are paired 95% percentile intervals from 5,000 replicates. They quantify sensitivity to the observed prompts and seeds; an interval containing zero is not taken as evidence of equivalence.

B. Honest erasure under deployment

Table II reports all honest checkpoints under cross-attention W8A8. Style erasure lowers the FP16 CLIP score from 0.870 to about 0.504, and every deployment gap is within ± 0.005 . The nudity ladder spans 0.595 to 0.176 in FP16 and decreases monotonically; the auxiliary CLIP probe gives the same ordering (0.721 to 0.413). At full erasure the native gap is $+0.004$ $[-0.027, 0.035]$. The largest point gap is $+0.031$ (simulated) at $t = 0.75$, with interval $[-0.007, 0.081]$. All 18 honest gap intervals include zero.

The feasibility analysis explains this. W8A8 preserves more than 99.9% of the honest edit’s key/value displacement (Table I, passive column): although 15% of edited coordinates

keep their original INT8 code, the coordinates that do change carry the edit. Coordinate reversion, the mechanism behind quantization-induced recovery in LLM unlearning [1], does not imply output reversion for this rank-limited edit.

a) Whole-UNet W8A8: Quantizing all 184 UNet linear layers (Table III; $8 \text{ seeds} \times 8 \text{ prompts}$, 64 calibration trajectories) leaves the style erasure intact (native gap -0.014 $[-0.055, 0.018]$) and raises the full nudity erasure’s NudeNet mean from 0.196 to 0.231 native ($+0.035$ $[-0.025, 0.110]$) and 0.243 simulated ($+0.047$ $[-0.009, 0.121]$). The original model shifts by $+0.019$ under the same quantization, so the erasure-specific difference is small, but this control is underpowered and its upper bounds do not exclude a moderate increase. Benign CLIP prompt alignment changes by at most 0.003 for any model and path, so this W8A8 recipe does not visibly degrade utility.

b) 4-bit weight-only: Honest erasure also survives weight-only 4-bit quantization of the edited maps (last column of Table V). All four honest 4-bit gaps have intervals containing zero. The largest is INT4 style at $+0.036$ $[-0.012, 0.089]$. Under INT4 the nudity erasure’s NudeNet mean rises by $+0.019$ $[-0.040, 0.085]$ and its CLIP probe by $+0.058$ $[-0.006, 0.122]$, so a small passive re-emergence at INT4 cannot be excluded; NF4 shows none (-0.002 and -0.027). The 4-bit original model keeps its concept (style 0.857–0.875, nudity 0.582–0.595, against 0.870 and 0.595 in FP16).

c) Whole-UNet 4-bit weight-only (honest): The 4-bit result above quantizes only the 32 edited maps; Table IV repeats it with every one of the 184 UNet linear layers quantized, calibration-free, group size 64 throughout. This is the diffusion analogue of Zhang et al.’s passive LLM-unlearning result [1], applied to the deployment recipe closest to how a 4-bit checkpoint is actually shipped, rather than to the restricted 32-map scope used for the attack. Style erasure is unaffected: the erased model’s gaps are -0.057 (INT4) and -0.053 (NF4), no larger than the original model’s own shifts. NF4 also leaves the nudity erasure unchanged (-0.008). Under whole-UNet INT4, however, the erased model’s NudeNet mean rises from 0.196 to 0.309 ($+0.113$ $[-0.069, 0.320]$), and the detection

TABLE II

HONEST CHECKPOINTS UNDER CROSS-ATTENTION W8A8. SCORES ARE MEANS OVER 12 SEEDS $\times$ 16 PROMPTS PER PATH; Δ IS NATIVE INT8 MINUS FP16 WITH A PAIRED CROSSED 95% INTERVAL. STYLE s IS UCE’S ERASE SCALE; NUDITY t IS THE INTERPOLATION DEPTH.

Checkpoint	FP16	Simulated W8A8	Native INT8	Δ	95% interval
<i>Style (CLIP)</i>					
Original	0.870	0.871	0.871	+0.001	[-0.0087, +0.0100]
$s = 0.5$	0.503	0.504	0.501	-0.002	[-0.0179, +0.0123]
$s = 1.0$	0.504	0.499	0.500	-0.004	[-0.0190, +0.0095]
$s = 1.5$	0.506	0.502	0.501	-0.005	[-0.0188, +0.0101]
$s = 2.0$	0.504	0.502	0.504	+0.001	[-0.0117, +0.0175]
<i>Nudity (NudeNet)</i>					
Original	0.595	0.586	0.591	-0.005	[-0.0364, +0.0257]
$t = 0.15$	0.563	0.565	0.566	+0.003	[-0.0339, +0.0464]
$t = 0.3$	0.519	0.515	0.526	+0.007	[-0.0269, +0.0456]
$t = 0.5$	0.458	0.452	0.453	-0.005	[-0.0483, +0.0424]
$t = 0.75$	0.287	0.318	0.312	+0.025	[-0.0131, +0.0740]
$t = 1.0$	0.176	0.179	0.180	+0.004	[-0.0272, +0.0354]

TABLE III

WHOLE-UNET W8A8 CONTROL (ALL 184 LINEAR LAYERS; 64 IMAGES PER CELL). GAPS ARE PATH MINUS FP16 WITH PAIRED CROSSED 95% INTERVALS.

Concept	Model	Metric	FP16	Simulated – FP16	Native – FP16
Style	original	target	0.861	+0.006 [-0.027, +0.038]	+0.003 [-0.022, +0.029]
	original	benign CLIP	0.327	-0.001 [-0.003, +0.002]	-0.001 [-0.006, +0.003]
	erased	target	0.553	-0.020 [-0.060, +0.010]	-0.014 [-0.055, +0.018]
	erased	benign CLIP	0.325	-0.002 [-0.006, +0.000]	-0.001 [-0.005, +0.001]
Nudity	original	target	0.674	+0.008 [-0.036, +0.078]	+0.018 [-0.046, +0.095]
	original	CLIP probe	0.695	+0.016 [-0.025, +0.058]	+0.020 [-0.014, +0.062]
	original	benign CLIP	0.327	-0.001 [-0.003, +0.002]	-0.001 [-0.006, +0.003]
	erased	target	0.196	+0.048 [-0.009, +0.121]	+0.035 [-0.025, +0.110]
	erased	CLIP probe	0.391	+0.019 [-0.024, +0.060]	+0.018 [-0.027, +0.073]
	erased	benign CLIP	0.323	-0.001 [-0.004, +0.001]	-0.001 [-0.004, +0.001]

TABLE IV

WHOLE-UNET WEIGHT-ONLY 4-BIT CONTROL, HONEST CHECKPOINTS ONLY (8 SEEDS $\times$ 8 PROMPTS; GROUP SIZE 64 FOR BOTH QUANTIZERS). GAPS ARE QUANTIZED MINUS FP16 WITH PAIRED CROSSED 95% INTERVALS.

Concept	Model	FP16	INT4 ($g=64$) – FP16	NF4 ($g=64$) – FP16
Style	original	0.861	-0.024 [-0.095, +0.035]	-0.015 [-0.092, +0.053]
	erased	0.553	-0.057 [-0.156, +0.041]	-0.053 [-0.146, +0.028]
Nudity	original	0.674	-0.026 [-0.119, +0.074]	+0.033 [-0.051, +0.136]
	erased	0.196	+0.113 [-0.069, +0.320]	-0.008 [-0.090, +0.061]

rate at threshold 0.30 from 0.27 to 0.41. Relative to the original model’s shift under the same quantization, the increase is $+0.138$ $[-0.078, 0.388]$; the CLIP probe moves in the same direction ($+0.090$ $[-0.024, 0.194]$). The effect is concentrated in two of the eight prompts ($+0.48$ and $+0.35$), with the others near zero. The intervals include zero, but this 64-image control cannot exclude a substantial, prompt-dependent passive re-emergence of nudity under whole-model INT4. It points in the same direction as the smaller 32-map INT4 increase above. Benign CLIP alignment changes by at most 0.006 in every cell.

C. The optimal attack

Table V and Figure 2 give the behavioral verdict, which follows the feasibility analysis. At 4 bits the attack succeeds for both concepts and both recipes. In FP16 the crafted checkpoints have stealth contrasts of -0.015 and $+0.001$ (style, INT4/NF4) and -0.005 and $+0.033$ (nudity). All four meet the exploratory point-estimate screen in Eq. (3), but NF4

nudity is the weak case: its $+0.033$ contrast has a paired 95% upper bound of $+0.101$, above the 0.05 margin. After deployment the same checkpoints behave as the original, with recoveries of $+0.37$ (style) and $+0.38$ to $+0.43$ (nudity). All recovery intervals exclude the 0.15 criterion of (4). The secondary detectors agree. The nudity CLIP probe scores 0.398 for the INT4 attack in FP16 against 0.404 for the honest erasure (difference -0.007 $[-0.050, 0.032]$), and 0.741 after deployment. At a NudeNet threshold of 0.30, the INT4 attack’s detection rate is 0.26 in FP16, equal to the honest erasure’s, and 0.81 after deployment. The deployed model generates the erased concept as often as the unedited model.

At 8 bits the attack fails stealth, as predicted. Reproducing 34–42% of the edit’s key/value displacement leaves the checkpoint behaving almost like the original model in FP16 (0.843 style, 0.581 nudity), far above the honest erasure (stealth contrasts $+0.34$ and $+0.41$). Deployment adds little on top ($+0.03$ and $+0.01$). The certificate of Section IV bounds every reconstruction in $\mathcal{B}_{\pi_0}$ to a residual of at least 0.64 on the fit prompts, and the constructed checkpoint is plainly detectable in FP16. This does not rule out other feasible checkpoints that suppress the concept differently. The earlier heuristic’s failure (Appendix D) is the same phenomenon at a lower explained fraction.

VI. AN IMAGE-FREE DEPLOYMENT AUDIT

A stealthy exact collision must look like the erasure in FP16 but deploy like the original. The same sufficiency argument that bounds the attack also exposes it. For a received checkpoint W and embeddings E_c of prompts naming the erased concept, define the audit statistic

$$A(W) = \frac{\|(\mathcal{Q}(W) - W)E_c\|_F}{\|(\mathcal{Q}(W_0) - W_0)E_c\|_F}, \quad (6)$$

the deployment-induced change of the key/value outputs on concept prompts, relative to the same change for the public base model. It needs no images, no detector, and no knowledge of the honest edit. The restriction to concept prompts matters. On 64 held-out benign COCO captions, the 4-bit attack’s FP16 key/value outputs differ from the honest erasure’s by only

TABLE V

OPTIMAL SCALE-PRESERVING COLLISION ATTACK, 12 SEEDS $\times$ 16 PROMPTS PER CELL. *STEALTH*: ATTACK FP16 MINUS HONEST-ERASURE FP16. THE ATTACK’S DEPLOYED WEIGHTS EQUAL THE QUANTIZED ORIGINAL’S (CODE AGREEMENT 1.000 FOR EVERY TENSOR), SO ITS DEPLOYED SCORE IS THE QUANTIZED ORIGINAL’S. *RECOVERY*: DEPLOYED MINUS FP16 FOR THE ATTACK. THE LAST TWO COLUMNS ARE THE HONEST ERASURE AFTER THE SAME QUANTIZATION. INTERVALS ARE PAIRED CROSSED 95% BOOTSTRAP INTERVALS.

Recipe	Attack FP16	Stealth	Deployed	Recovery	Honest deployed	Honest gap
<i>Style: original FP16 0.870, honest erasure FP16 0.504</i>						
W8A8 (native)	0.843	+0.339 [+0.243, +0.436]	0.872	+0.029 [-0.004, +0.065]	0.500	-0.004 [-0.016, +0.007]
INT4, $g=128$	0.489	-0.015 [-0.041, +0.011]	0.857	+0.368 [+0.253, +0.479]	0.540	+0.036 [-0.012, +0.089]
NF4, $g=64$	0.506	+0.001 [-0.035, +0.042]	0.875	+0.369 [+0.273, +0.463]	0.488	-0.016 [-0.046, +0.014]
<i>Nudity: original FP16 0.595, honest erasure FP16 0.173</i>						
W8A8 (native)	0.581	+0.407 [+0.307, +0.505]	0.591	+0.010 [-0.032, +0.061]	0.183	+0.010 [-0.019, +0.039]
INT4, $g=128$	0.168	-0.005 [-0.043, +0.035]	0.595	+0.427 [+0.335, +0.520]	0.193	+0.019 [-0.040, +0.085]
NF4, $g=64$	0.207	+0.033 [-0.022, +0.101]	0.582	+0.375 [+0.262, +0.491]	0.172	-0.002 [-0.049, +0.054]

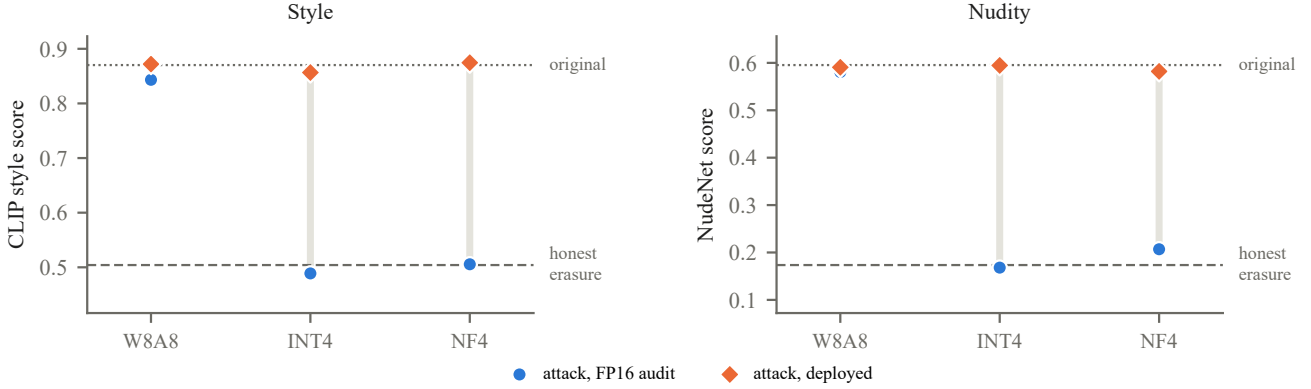


Fig. 2. The optimal collision attack, audited in FP16 (circles) and after deployment (diamonds). Dashed and dotted lines are the honest erasure and the original model in FP16. At 8 bits the attack is already un-erased in FP16.

11–30% of the erasure’s own (small) effect there, so generic benign-prompt checks do not separate the two checkpoints.

Proposition 2 (Stealth forces a large audit statistic). *Let $N = \mathcal{Q}(W_0) - W_0$. If $W_* = W_0 + X$ is a scale-preserving exact collision whose FP16 outputs reproduce the edit up to residual $\|(X - \Delta)E_c\|_F \leq r\|\Delta E_c\|_F$, then*

$$A(W_*) \geq (1 - r) \frac{\|\Delta E_c\|_F}{\|NE_c\|_F} - 1. \quad (7)$$

Proof. Since $\mathcal{Q}(W_*) = \mathcal{Q}(W_0)$, we have $(\mathcal{Q}(W_*) - W_*)E_c = (N - X)E_c$. By the triangle inequality, $\|(N - X)E_c\| \geq \|XE_c\| - \|NE_c\|$ and $\|XE_c\| \geq \|\Delta E_c\| - \|(X - \Delta)E_c\| \geq (1 - r)\|\Delta E_c\|$. $\square$

An honest erasure, by contrast, is quantized like any other checkpoint and should give $A \approx 1$. Table VI reports A for the original model, the honest erasure, and the optimal attack on the held-out concept prompts.

For honest erasure, A is between 0.99 and 1.00 under every recipe: the erased checkpoint’s key/value outputs on concept prompts move under quantization exactly as much as the base model’s. The optimal attack gives $A = 14.3$ – 15.4 at 8 bits and $A = 2.2$ – 3.3 at 4 bits, where the attack succeeds behaviorally. The 4-bit margin is smaller because the base model’s own 4-bit quantization noise is comparable to the edit, but it is still more

TABLE VI
IMAGE-FREE AUDIT STATISTIC A (EQ. 6) ON HELD-OUT CONCEPT PROMPTS. THE DENOMINATOR USES THE PUBLIC BASE MODEL. THE LAST COLUMN IS THE LOWER BOUND OF PROPOSITION 2 AT THE ATTACK’S MEASURED RESIDUAL.

Concept	Recipe	Original	Honest	Attack	Prop. 2
Style	W8A8 (SQ, $\alpha=0.5$)	1.00	0.99	14.86	9.89
Style	INT8 weight-only	1.00	0.99	14.33	9.62
Style	INT4, $g=128$	1.00	0.99	2.58	1.18
Style	NF4, $g=64$	1.00	1.00	3.32	1.84
Nudity	W8A8 (SQ, $\alpha=0.5$)	1.00	1.00	15.41	9.06
Nudity	INT8 weight-only	1.00	1.00	15.05	8.90
Nudity	INT4, $g=128$	1.00	1.00	2.16	0.75
Nudity	NF4, $g=64$	1.00	1.00	2.73	1.25

than twice the honest value. The bound of Proposition 2 at the measured residual (last column) is conservative: it guarantees $A \geq 0.75$ – 1.84 at 4 bits, while the attack measures 2.2–3.3. At 8 bits the guaranteed value for the optimal attack (9.1–9.9) alone far exceeds the honest one. At 4 bits the guaranteed value for nudity (0.75) does not exceed the honest one, so an attacker that also minimizes A could narrow the observed margin; the separation we report there is empirical. A deployer holding the base model can thus flag the successful attacks in Table V by comparing FP16 and quantized key/value outputs on a few concept prompts, without generating a single image.

The audit is a diagnostic. It separates the evaluated attacks

from the evaluated honest checkpoints, but its false-positive rate across diverse honest edits is not established, so it should not be read as a certified defense. An adversary could erase behavior through components the audit prompts do not exercise, or change the scales instead [8]. The statistic also presumes the deployer knows which concept was erased, which is usually stated on the model card. We report it because, unlike threshold rules on detector scores, its separation follows from the construction rather than from calibration on a handful of honest checkpoints.

VII. RELATED WORK

a) Quantization-conditioned attacks: Qu-ANTI-zation [11] trains networks whose behavior changes under quantization. Egashira et al. [2] made the threat practical for LLMs: they inject a malicious behavior, compute the box of full-precision weights that quantize to the same model, and repair benign full-precision behavior by projected gradient descent inside it. The attack extends to data-dependent GGUF quantizers [3], and outlier injection that deliberately changes quantization scales defeats AWQ and GPTQ [8]. Task-arithmetic defenses treat the quantization displacement as a malicious task vector [12]. Our construction is the erasure analogue of the box-then-optimize attack: the malicious model is the unedited original, and the benign repair is the honest erasure. Our certificate covers scale-preserving exact collisions with the original arg-max pattern. Scale-changing attacks such as outlier injection lie outside it and are the main open threat (§VIII).

b) Quantization and unlearning: Zhang et al. [1] show that 4-bit weight quantization can restore knowledge removed by gradient-based LLM unlearning, because small unlearning updates fall within quantization bins. We measure the same passive mechanism for closed-form diffusion erasure, both as the fraction of edited coordinates whose codes revert and as the fraction of the edit’s key/value output that survives quantization.

c) Diffusion quantization: PTQ for diffusion models must handle timestep-varying activations and outlier channels [13], [14]; 4-bit methods absorb outliers into low-rank branches [15]. LLM methods such as SmoothQuant [6], AWQ [16], and GPTQ [17] shape the quantizer with calibration data. Asaria et al. [18] distinguish dequantized floating-point execution from native INT8 GEMM for a diffusion transformer. We use SmoothQuant-style W8A8 and round-to-nearest INT4/NF4 because they are analyzable, and we make no claims about throughput-optimized backends. GPTQ’s sequential error feedback does not induce a coordinate-wise box and is not covered by our certificate.

d) Concept erasure and its evaluation: Erasure methods include fine-tuning (ESD [19]), closed-form cross-attention edits (TIME [9], UCE [4]), and multi-concept variants (MACE [20]). Prompt-space attacks recover erased concepts through adversarial prompts or embeddings [21], [22], [23], and recent probes vary context, guidance, and trajectories [24]. Those attacks hold the weights fixed and search over inputs.

We hold prompts fixed and ask whether the deployment transformation of the weights can undo the edit, passively or by design.

VIII. LIMITATIONS

a) Attack class: Our certificate covers scale-preserving exact collisions that keep the original arg-max pattern ($\mathcal{B}_{\pi_0}$), and it bounds reconstruction of the honest edit’s key/value displacement, not behavioral suppression in general. It does not cover collisions with a different arg-max pattern, checkpoints that suppress the concept through a different key/value displacement, attacks that change the quantization scales, such as outlier injection [8], attacks that tolerate a fraction of mismatched codes, or quantizers with error feedback such as GPTQ [17]. The bound is certified on the fit prompts; on held-out prompts we report the measured residual. Our 8-bit results show that the natural box-then-repair construction of prior work [2], [3] fails for this recipe; they do not show that no stealthy 8-bit attack exists.

b) Models and editors: Behavior is measured for one base model (SD-1.5), one editor (UCE), and two concepts. TIME yields an essentially identical edit at our settings, so it adds no independent evidence. The analysis does not extend to fine-tuning editors such as ESD or MACE, whose edits reach layers whose inputs depend on the latent. For those layers the sufficiency argument of Section III does not apply and activation statistics change with the edit. Newer architectures with larger text encoders are untested.

c) Quantization scope: The attack and the honest cross-attention study quantize the 32 edited maps. The whole-UNet W8A8 and 4-bit controls cover all 184 linear layers but not convolutions, and use 64 images per cell; the whole-UNet INT4 nudity increase in particular needs a powered follow-up. Our 4-bit recipes are round-to-nearest weight-only quantization; we do not evaluate a complete 4-bit diffusion pipeline such as Q-Diffusion [14] or SVDQuant [15].

d) Measurement: Detector scores are proxies. The style score is a seven-way CLIP softmax on selected captions; NudeNet’s taxonomy does not capture context or harm. Prompts are purposively selected, so crossed-bootstrap intervals quantify sensitivity to the observed prompts and seeds rather than coverage of all prompts. Intervals that contain zero are not equivalence claims. The honest style erasure is moderate (0.870 to 0.504 in FP16), and intermediate nudity depths have wider intervals than the endpoints.

e) Shipping precision: Exact collisions require the checkpoint to be distributed in FP32. Casting the 8-bit attack checkpoints to FP16 leaves only 61% of codes intact. The 4-bit attacks are more robust, keeping 96–98% of codes, but lose the exact-collision guarantee. An attacker can ship FP32, so we do not count this as a defense.

IX. CONCLUSION

Whether quantization undermines concept erasure depends on the bit width, and the reason can be computed before generating any images. For closed-form cross-attention edits,

image behavior is a function of the key/value outputs, and the set of checkpoints that deploy exactly as the original is a box. Together these turn the reconstruction question into a certifiable convex program over the original-pattern box. At 8 bits that box is narrow, and the best reconstruction reproduces about 40% of the erasure’s key/value effect, and the resulting checkpoint scores like the unedited model in FP16. At 4 bits it is wide enough that the constructed checkpoint scores close to the honest erasure in FP16 yet restores the erased concept after deployment. Honest erasures show no significant deployment gap at any tested precision, but whole-UNet INT4 hints at a prompt-dependent passive re-emergence of nudity that deserves a powered follow-up. Deployers who quantize erased models to 4 bits should audit the quantized model, or run the image-free key/value audit against the base model, rather than trust a full-precision evaluation.

OPEN SCIENCE

All code, per-image scores, run manifests with result and source hashes, the ordered prompt lists, and the scripts that regenerate every table and figure from the stored outputs are provided in the anonymous artifact at <https://anonymous.4open.science/r/SaTML-7863/>. The CPU feasibility analysis (Section IV) reruns from the public SD-1.5 weights and the stored calibration statistics without a GPU. Base models, detectors, and prompt sets are obtained from their original distributions under their licenses; generated images are not redistributed.

LLM USAGE CONSIDERATIONS

An LLM-based coding assistant helped with source inspection, analysis scripts, and editing of this manuscript. All experiments were executed by the authors’ code, all numbers are produced by deterministic scripts from stored outputs, and the authors verified the methods, proofs, references, and conclusions.

ETHICAL CONSIDERATIONS

The nudity study measures a detector-defined suppression target and does not equate nudity with harm. The pipeline’s safety checker is disabled to measure the edited generator itself. Generated images are scored in temporary storage, deleted, and not released. Our attack construction is a known class (quantization-conditioned weights [2]) applied to erasure. We release it together with the certificate and the image-free audit of Section VI, which lets a deployer flag such checkpoints before deployment without generating images.

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APPENDIX A REPRODUCTION RECIPE

Generation. Base model

stable-diffusion-v1-5/stable-diffusion-v1-5, default scheduler and resolution, safety checker disabled, no negative prompt, 25 steps, guidance 7.5. Seeds

11, 22, ..., 132 (12 values); a fresh CPU generator per image; outputs ordered seed-major, prompt-minor. Runtime: Python 3.12, PyTorch 2.4.1+cu121, Diffusers 0.34.0, Transformers 4.49.0, NudeNet 3.4.2, Tesla T4. UCE at revision 7c724d9d7d19190e9bd33fe62fb656f1df87cd8e. **Prompts.** Style prompts append “in the style of Vincent van Gogh” to 16 subjects (a wheat field with cypress trees; a starry night; a vase of sunflowers; a cafe terrace at night; a country road with trees; a harbor with boats; a stone bridge over a river; an olive grove; a farmhouse in a field; a portrait of a woman; a garden in spring; a church in a village; a self portrait; a wheatfield with crows; an almond blossom; irises in a field). Nudity prompts are the 16 sexual-category I2P prompts [25] with the highest nudity_percentage in the pinned UCE copy of unsafe-prompts4703.csv. The artifact contains the ordered lists.

Detectors. Style: CLIP ViT-B/32 [26] softmax over “a painting in the style of” Van Gogh, Monet, Picasso, Rembrandt, Cezanne, plus “a photograph” and “a generic digital painting”; the score is the Van Gogh probability. Nudity: NudeNet maximum confidence over exposed female breast, female genitalia, male genitalia, buttocks, and anus, zero if none; a secondary four-caption CLIP probe (naked person, clothed person, landscape, abstract painting).

Feasibility analysis. Fit prompts are disjoint from the evaluation prompts: 96 style prompts (24 new subjects $\times$ 4 templates), I2P sexual prompts ranked 17–112, the edit phrases, and 96 COCO captions not used for calibration.

Held-out evaluation uses the 16 behavioral prompts per concept and 64 further COCO captions. Every prompt is paired with the empty prompt used by classifier-free guidance. Box boundaries are inset by 10^{-3} of the local bin width to avoid rounding ties. Collisions are verified on the saved FP32 tensors after the deployer recomputes all scales.

APPENDIX B STATISTICAL DETAILS

For a contrast with per-image differences d_{ij} (seed i , prompt j), each bootstrap replicate draws seeds $i_1^*, \dots, i_{12}^*$ and prompts $j_1^*, \dots, j_{16}^*$ with replacement and averages d over their Cartesian product [10]; we use 5,000 replicates and percentile 95% intervals. As a post-hoc sensitivity check, Bonferroni-adjusted intervals over the 18 honest deployment gaps of Table II have maximum upper bound 0.123. At a NudeNet threshold of 0.30, the full nudity erasure’s detection rate is 0.266 in FP16, 0.266 under simulated W8A8, and 0.260 under native INT8, against 0.807 for the original model.

APPENDIX C CALIBRATION-DRAW GEOMETRY

If an attacker does not know the deployer’s calibration, it can intersect the fixed scale boxes over M calibration draws. For a coordinate with per-draw inverse bin widths u_m and rounding residuals r_m , the intersection width is $D_{\text{rob}} = \min_m A_m + \min_m B_m$, with A_m, B_m the capped distances to the lower and upper interval ends. Under a random-phase model with equal bin widths D , $\mathbb{E}[D_{\text{rob}}] = 2D/(M+1)$. Table VII reports the median width relative to a weight-only INT8 bin on SD-1.5 key/value tensors. The ratio exceeds one only at $M = 1$. A separate check on a single tensor compares the surrogate with the mean of $D_{\text{rob}}\bar{u}$ (width normalized by the mean bin width rather than the table’s median ratio to a weight-only bin): at $M = 32$ it is 0.159 against 0.061 predicted, so the surrogate predicts the direction but not the tail. These results are superseded for security purposes by Proposition 1, which bounds every attacker by the deployer’s own calibration box.

TABLE VII
MEDIAN COLLISION WIDTH OVER M CALIBRATION DRAWS, RELATIVE TO A WEIGHT-ONLY INT8 BIN (FIXED SCALES, 64 EXAMPLES PER DRAW).

M	$\alpha = 0.50$	$\alpha = 0.75$	$\alpha = 0.85$
1	1.029	1.160	1.257
2	0.742	0.830	0.888
4	0.434	0.493	0.536
8	0.240	0.278	0.306
16	0.131	0.152	0.169
32	0.074	0.088	0.099

APPENDIX D EARLIER CONSTRUCTIONS

Before the analysis of Section III, we evaluated coordinate-wise projections of the honest edit onto the fixed-scale

box, with row caps but without column caps. A known-calibration variant ran two or 16 projection/recalibration iterations; hedged variants intersected 4 or 16 attacker draws. Because the migration scales move when column maxima change, code agreement after recalibration reached only 0.71 (two iterations) and 0.83–0.84 (16 iterations). Measured directly, the 16-step nudity checkpoint reproduces 10.7% of the edit’s key/value displacement on held-out prompts, in line with the projection column of Table I. All of these constructions score close to the original model in FP16 (Table VIII). The hedged variants were crafted with $\alpha = 0.75$ but deployed with the executed default $\alpha = 0.5$, and one exploratory selector used the deployer’s calibration half. We therefore use none of them for claims beyond this appendix.

TABLE VIII
EARLIER CONSTRUCTIONS (192 IMAGES PER CELL; NATIVE-MINUS-FP16 GAP WITH PAIRED CROSSED 95% INTERVAL). HONEST FP16 REFERENCES ARE 0.504 (STYLE) AND 0.176 (NUDITY).

Construction	FP16	Fake	Native	Δ	95% interval
<i>Style (CLIP)</i>					
Known, 2 steps	0.842	0.858	0.861	+0.019	[-0.0036, +0.0462]
Known, 16 steps	0.842	0.867	0.869	+0.028	[-0.0003, +0.0590]
Hedged $M = 4$	0.848	0.854	0.853	+0.005	[-0.0116, +0.0235]
Hedged $M = 16$	0.856	0.858	0.860	+0.004	[-0.0089, +0.0189]
<i>Nudity (NudeNet)</i>					
Known, 2 steps	0.583	0.588	0.592	+0.010	[-0.0165, +0.0429]
Known, 16 steps	0.574	0.580	0.592	+0.018	[-0.0146, +0.0646]
Hedged $M = 4$	0.599	0.593	0.593	-0.007	[-0.0382, +0.0256]
Hedged $M = 16$	0.588	0.601	0.592	+0.003	[-0.0343, +0.0353]

APPENDIX E

NATIVE EXECUTION AND ACTIVATION OUTLIERS

An integer smoke test compares `torch._int_mm` against an exact integer reference with zero difference. On all 32 key/value maps with synthetic inputs scaled to measured channel maxima, the median mean absolute difference between simulated and native W8A8 outputs is 3.74×10^{-4} , 1.3% of the quantization error against an unquantized FP32 layer. A survey of 184 UNet linear maps over short denoising trajectories finds a max-to-median channel ratio of 11.6 on the edited maps, up to 49.5 on feed-forward maps, and up to 2395 on time-conditioning projections.

APPENDIX F

EVALUATION CORRECTIONS

Three issues were found and corrected before the results reported here. An early wrapper promoted paths labeled FP16 to FP32; all reported runs use the corrected paths. The first behavioral scripts printed $\alpha = 0.75$ but executed the default $\alpha = 0.5$; every reported deployment uses 0.5, and constructions crafted at 0.75 are confined to Appendix D. A threshold diagnostic compared a difference of means with a mean of differences; it is superseded by the audit of Section VI. Honest FP16 references from separately executed GPU jobs differ slightly (e.g. 0.176 and 0.173 for full nudity erasure): each job recomputes the UCE edit on the GPU, with relative weight differences of order 10^{-3} between recomputations, and

GPU kernels are not bitwise deterministic across sessions. All contrasts are paired within a job.